

Requirements Traceability Matrix (RTM)

HealthFirst Care – Patient Experience & Operational Efficiency Initiative

1. Project Context

This RTM builds on the Business Requirements Document (BRD) and Stakeholder Profiles produced in the prior lab. HealthFirst Care faces three recurring, interrelated challenges, reported consistently across stakeholder groups:

- **Long patient wait times** — patients report waits of 30–60 minutes past scheduled times with no updates during delays.
- **Resource allocation inefficiencies** — doctors and nurses report overbooked schedules, limited diagnostic equipment during peak hours, uneven nurse-to-patient ratios, and supply shortages.
- **Communication gaps across departments** — delayed lab/radiology results, poor handoff coordination, double bookings from limited visibility into doctor availability, and unintegrated scheduling/records systems (per IT).

The Requirement ID, Priority, Stakeholder(s), and Project objective columns of the RTM (Section 3) map each requirement back to these challenges and to the BRD's five business objectives: reducing wait times, reducing double bookings, improving inter-departmental communication, improving patient satisfaction, and eliminating data silos within budget.

2. Requirement Categorization & MoSCoW Rationale

Requirements from the BRD were split into Functional (FR) and Non-Functional (NFR) categories, plus four out-of-scope items carried from the BRD's "Won't-have" list (WH1–WH4) so the prioritization set is complete. Rationale for each MoSCoW rating:

ID	Category	Priority	Why
FR1	Functional	Must Have	Directly targets double bookings named by the Appointment Scheduler; core to the wait-time objective.
FR2	Functional	Must Have	Addresses the most common patient complaint (no updates during delays).
FR3	Functional	Must Have	Resource data shows recurring unavailability; needed for FR1 to avoid conflicts.
FR4	Functional	Should Have	Valuable for cross-department coordination, but the system can deliver value without it initially.
FR5	Functional	Must Have	IT-flagged precondition for accurate, non-duplicated availability data.
FR6	Functional	Should Have	Needed to prove objectives are being met, but not required for core operation.
NFR1	Non-Functional	Must Have	Poor usability would undermine adoption of FR1/FR2.
NFR2	Non-Functional	Must Have	IT reports frequent downtime already; reliability is foundational.
NFR3	Non-Functional	Must Have	Legacy security vulnerabilities are non-negotiable when handling patient data.
NFR4	Non-Functional	Should Have	Improves the peak-time experience but can be tuned post-launch.
NFR5	Non-Functional	Could Have	Forward-looking scalability recommendation, not required at current volumes.

WH1–WH4	Out of scope	Won't Have	Excluded explicitly by the BRD's budget/scope constraints.
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3. Data File Findings

Findings below are exact counts/averages computed from the three provided CSV files and are used in the RTM's "Related data file" column to justify specific requirements. Each file is a small sample (10–20 rows); figures are cited by file name so they can be independently re-checked.

Data file	Key finding	Requirement(s) justified
Appointment data (n=20)	Only 8/20 appointments (40%) were Completed; No Show, Rescheduled, and Cancelled each account for 4/20 (20%). Each department runs a single fixed time slot (e.g., Cardiology 10:00 AM), pointing to scheduling rigidity and conflicts.	FR1 (automated scheduling/conflict detection); NFR4 (peak-time performance)
Feedback data (n=15)	Overall average satisfaction is 7.0/10. General Medicine (5.33) and Paediatrics (6.0) score lowest. 6/15 responses (40%) scored ≤ 6 , with comments citing "Long wait times," "Delayed response," and "Room cleanliness issue."	FR2 (real-time notifications); NFR1 (usability)
Resource data (n=10)	Nurse entries show 0 usage hours under "Unavailable" status (2 of 10 records); Room entries show 2 usage hours under "Under Maintenance" (2 of 10); Equipment shows 6 hours "In Use" (2 of 10); Doctor/Technician show 8–10 hours "Available" (4 of 10).	FR3 (real-time resource visibility)

4. Conclusion

The RTM links every requirement from the BRD to its originating stakeholder, its supporting project objective, the data evidence (where available) that justifies it, and a proposed priority and status. Must-have items concentrate on the scheduling, notification, resource-visibility, and security/reliability requirements most directly tied to the wait-time, double-booking, and communication problems documented in the BRD and confirmed by the appointment, feedback, and resource data reviewed above.